# FINISHED PIPE WALL

Measurement and internal-pressure thickness review

ASME B31.3-2018 | Straight pipe | Internal pressure

### DESIGN EQUATION

ASME B31.3-2018, 304.1.1 / 304.1.2, (2), (3a)
t = P D / [2 (S E W + P Y)]
tm = t + c; c = corrosion + erosion + mechanical

P and S in MPa; dimensions in mm. Input pressure in bar x 0.1 = MPa.

MINIMUM MEASURED WALL AFTER FABRICATION

## 3.4787 mm

**BELOW the internal-pressure wall requirement**

User-entered inspection minimum | Not calculated from Schedule

| | |
|---|---|
| Required remaining wall | **3.4787 mm** |
| Measured minus required | **-5.925e-07 mm** |

Required remaining wall = pressure thickness + corrosion + erosion. The measured value is the minimum remaining metal after ALL fabrication and machining, including groove/thread roots. It is supplied by the user; the app does not measure the pipe.

No mill under-tolerance or already removed metal is deducted from this measurement again.

### 01 PIPE AND INSPECTION

| | |
|---|---|
| Material / grade | **Example steel - verify actual material grade** |
| Pipe size / outside diameter | **4 inch / 114.3 mm** |

| | |
|---|---|
| Inspection / pipe reference | مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890مرجع فحص الماسورة بعد التصنيع - PIPE-ABC1234567890 |
| Project | **Example only - مثال توضيحي** |
| Prepared by | **QA preview - not a real inspection** |
| Report date | **2026-10-06 10:28** |

## 02 MEASUREMENT BASIS

Use an inspection minimum that accounts for measurement uncertainty. Exclude coatings, lining and weld reinforcement. This comparison applies to a new straight pipe before service. An average reading or nominal schedule wall is not a minimum finished-wall measurement.

MEETS / BELOW refers only to the internal-pressure wall requirement for these inputs. It is not an inspection certificate or approval of the complete piping design.

# DESIGN BASIS

Calculation trail and nominal pipe selection

## 03 PRESSURE DESIGN INPUTS

| | |
|---|---|
| Internal design gauge pressure P | **20 bar(g) = 2 MPa** |
| Design metal temperature | **20 °C** |
| Allowable stress S | **100 MPa** |
| Quality factor E / weld factor W | **0.85 / 0.9** |
| Coefficient Y / y | **0.4** |
| Corrosion / erosion allowance | **1.5 / 0.5 mm** |
| Mechanical removal allowance | **1 mm (before machining)** |
| Code / material input reference | **Illustrative coefficients only; not project design values.** |

## 04 CALCULATION TRACE

P is converted from bar to MPa; D and thicknesses are in mm. S is in MPa.

| | |
|---|---|
| Pressure thickness t | **$P \times D / [2 \times (S \times E \times W + P \times Y)] = 1.4787$ mm** |
| Required remaining finished wall | **t + corrosion + erosion = 3.4787 mm** |
| Required wall before machining tm | **t + corrosion + erosion + mechanical = 4.4787 mm** |
| Equation scope checks | **$t < D/6$; $P/(S \times E) = 0.0235 \leq 0.385$** |

## 05 NOMINAL PIPE ORDERING CHECK (SEPARATE)

| | |
|---|---|
| Specified minus tolerance | **12.5 %** |
| Minimum nominal pipe wall to order | **tm / (1 - tolerance/100) = 5.1185 mm** |
| Selected nominal pipe | **Sch 40 / 6.02 mm** |
| Lower wall bound from tolerance | **5.2675 mm - calculated bound, NOT measured** |
| Nominal selection result | **Meets the ordering thickness check** |

These are pipe wall dimensions. No flat-sheet thickness or forming-process calculation is performed. A nominal-wall check cannot replace the measured finished-wall result at the start of this report.

## 06 LIMITS AND REFERENCES

Basis: B31.3-2018, 304.1.1 and 304.1.2, equations (2) and (3a); Y: Table 304.1.1. Nominal dimensions: B36.10-2022 Table 2-1. S, E, W, allowances, tolerance and inspection values require verification against project specifications. This report does not assess external pressure, bends, branches, other loads, fatigue, remaining life or high-pressure special services. Comparisons use unrounded numbers; displayed numbers are formatted for readability.